

University of California Riverside
Electrical and Computer Engineering Department

Electromagnetics I

Lecture 3: Maxwell's Equations in Integral Form

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Office Hours: Thursday, 8:00 AM - 9:00 AM, MSE 016

Outline

- Maxwell's equations

Gauss's law for electric fields

Gauss's law for magnetic fields

Faraday's law

Ampere's law

- Physical units / Mathematical operators
- Lorentz Force
- Examples of using Maxwell's equations
- Homework #1

Chapter 1

Maxwell's Equations

Maxwell's equations are a set of partial differential equations that, together with the Lorentz force law, form the foundation of classical electromagnetism, classical optics, and electric circuits (Wikipedia).

Maxwell's equations are the mathematical relations between the electric and electric fields, and their associated charge and current distributions (Magdy F. Iskander, "Electromagnetic Fields and Waves", page 58.

Maxwell's equations are the following:

- 1) Gauss's law for electric field
- 2) Gauss's law for magnetic field
- 3) Faraday's law
- 4) Ampere's law

Maxwell's equations can be written in different forms (integral or differential) for different regions (e.g., vacuum or material region) and for static and time-varying fields.

In the textbook "Electromagnetic Fields and Waves", by Magdy F. Iskander,:
Maxwell's equation:

In differential forms, 141-147

In integral form, 58-70

In material regions, 208-11

For static fields, 76. 274-75

For time varying fields, 76-82

Maxwell's Equations

INTEGRAL FORMS

$$\oint_s \epsilon_o \epsilon_r \mathbf{E} \cdot d\mathbf{s} = \int_v \rho_v dv$$

$$\oint_s \mathbf{B} \cdot d\mathbf{s} = 0$$

$$\oint_c \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d}{dt} \int_s \mathbf{B} \cdot d\mathbf{s}$$

$$\oint_c \mathbf{H} \cdot d\boldsymbol{\ell} = \int_s \mathbf{J} \cdot d\mathbf{s} + \frac{d}{dt} \int_s \epsilon_o \epsilon_r \mathbf{E} \cdot d\mathbf{s}$$

DIFFERENTIAL FORMS

$$\boldsymbol{\nabla} \cdot \epsilon_o \epsilon_r \mathbf{E} = \rho_v \quad (4.1)$$

$$\boldsymbol{\nabla} \cdot \mathbf{B} = 0 \quad (4.2)$$

$$\boldsymbol{\nabla} \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (4.3)$$

$$\boldsymbol{\nabla} \times \mathbf{H} = \mathbf{J} + \frac{\partial \epsilon_o \epsilon_r \mathbf{E}}{\partial t} \quad (4.4)$$

$$\oint_S \epsilon_0 \epsilon_r \mathbf{E} \cdot d\mathbf{s} = \int_V \rho_v dv$$

Gauss' Law

free space
 $\epsilon_r = 1$

Total electric flux emanating from a closed surface equals the electric charge enclosed by that surface

$\vec{E}$ - electric field intensity
 (force/charge), units [V/m]

$$\epsilon_0 \oint_S \vec{E} \cdot d\vec{S} = Q$$

ρ_v - volume charge density
 (Q/volume), units [C/m³]

$\epsilon_0 = 8.85 \times 10^{-12}$ [F/m]
 dielectric constant,
 permittivity of vacuum

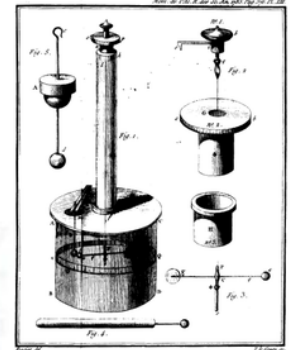
$$\epsilon_0 \oint_S \vec{E} \cdot d\vec{S} = \int_V \rho_v dv$$

Coulomb's Law

$$\mathbf{F} = k \frac{Q_1 Q_2}{R^2} \mathbf{a}_{12}$$



Charles-Augustin de Coulomb



Coulomb's torsion balance

The electric force between two point charges is directly proportional to the product of the charges and inversely proportional to the square of the distance between them.

$$k = \frac{1}{4\pi\epsilon_0}$$

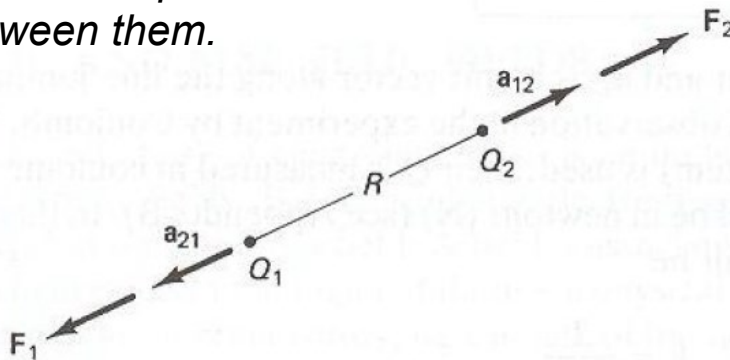


Figure 1.22 The electric force between two point charges Q_1 and Q_2 .

where ϵ_0 is called the permittivity of air (vacuum) and has a value measured in farads per meter (F/m),

$$\epsilon_0 = 8.854 \times 10^{-12} \approx \frac{1}{36\pi} \times 10^{-9} \text{ F/m}$$

Electric Field (Intensity) **E**

The electric field intensity **E** is the vector force on a unit positive test charge

Electric field units:
[N/C] or [V/m]

$$\vec{E} = \frac{\vec{F}}{q}$$

Force unit:
Newton

Charge units:
Coulombs

Units for Electric Charge

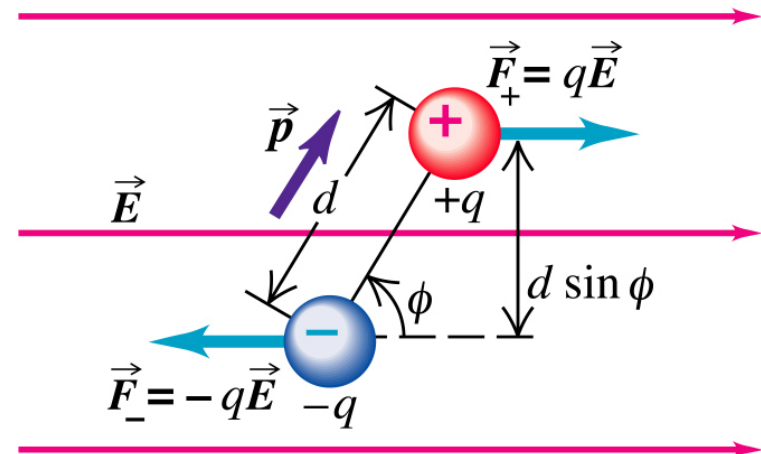
- A Coulomb is defined as the number of electrons passing through a current of 1 Amp each second:

$$C = A \cdot s.$$

- The conversion between C and e was determined by Robert A. Millikan:

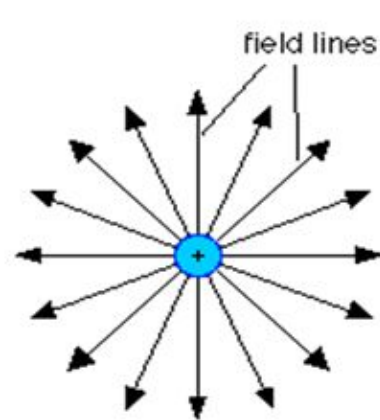
$$1 e = 1.60 \times 10^{-19} C.$$

- It is much easier to measure charge flowing in a current than to count protons and electrons individually!

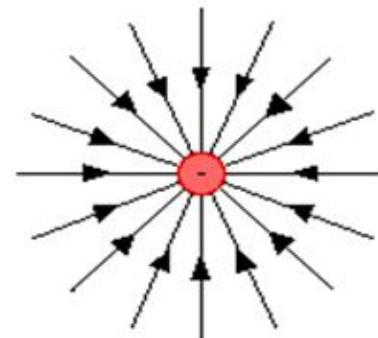


Electric Field Lines

- Electric fields are represented by electric field lines (*imaginary*) and show the direction of the force at that point.
- Electric field direction is from a positive charge to a negative charge.
- Electric field lines (also known as electric flux) do not touch each other.
- Electric field line represents the path a small positive charge would move in the electric field.



The electric field from an isolated positive charge

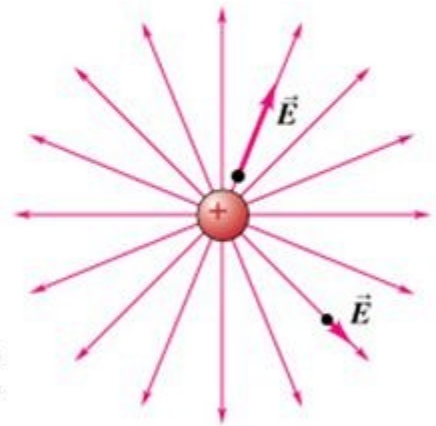


The electric field from an isolated negative charge

Electric field equation for a point charge

$$\vec{E} = \frac{\vec{F}_E}{q_0} \text{ becomes } E = k_c \frac{q}{r^2} \text{ for a point charge}$$

- E is electric field strength, units: N/C
- E is a VECTOR
- q_0 is a + test charge, units: C
- q is the charge producing the field, units: C
- r is the distance between charges, Units: m
- F_E is the Electric Force, units: N
- k_c is the coulomb constant, $8.99 \times 10^9 \text{ Nm}^2/\text{C}^2$

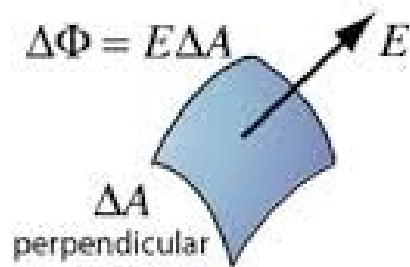


Electric Flux

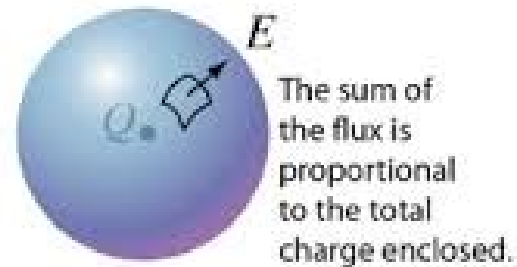
$$\epsilon_0 \oint_S \vec{E} \cdot d\vec{S} = Q$$

$$\phi_E = \mathbf{E} \cdot \mathbf{S} = ES \cos \theta$$

Electric flux is the measure of the electric field through a given surface



$$\Phi_{electric} = \frac{Q}{\epsilon_0}$$



Gauss's law for electric field

$$\epsilon_0 \oint_S \vec{E} \cdot d\vec{S} = \int_V \rho_v dV$$

closed surface integral

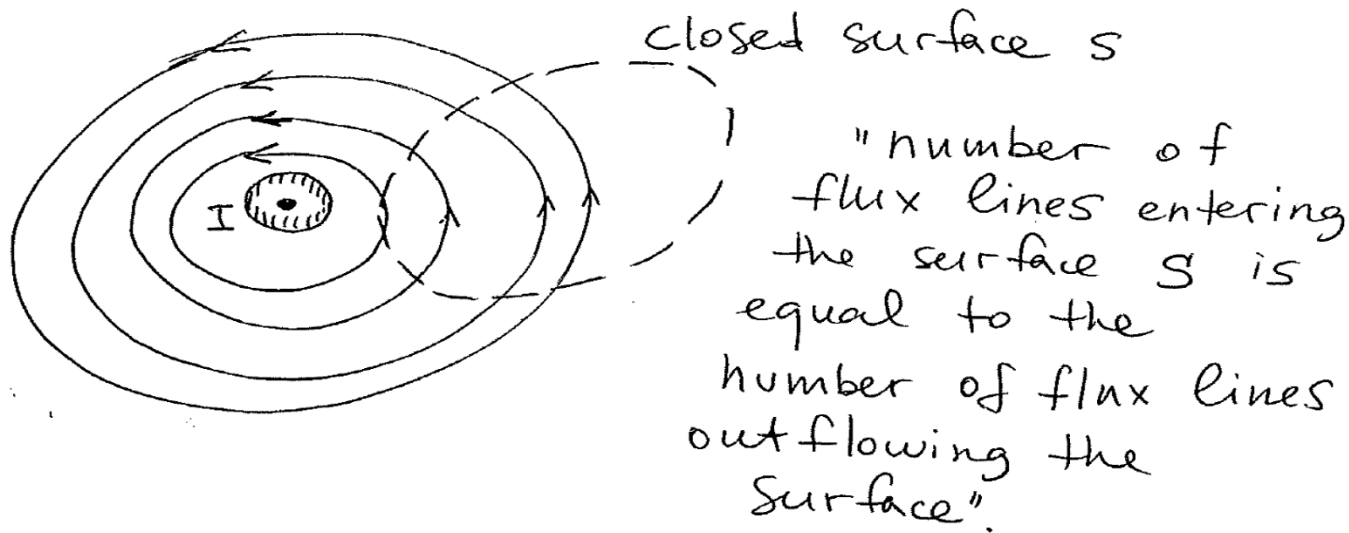
volume integral



Gauss' Law for Magnetic Field

$$\oint_S \vec{B} \cdot d\vec{S} = 0$$

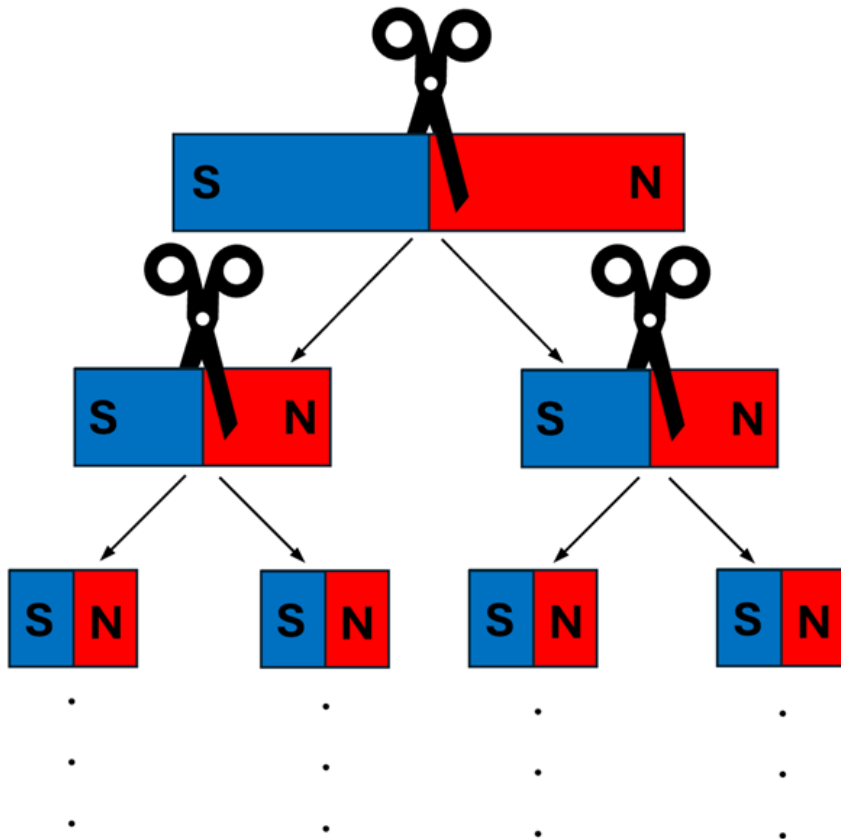
Total magnetic flux through the closed surface should be equal to zero



Carl Friedrich Gauss



The experiment looks so simple



Researcher	Approx. Year	Experimental Approach	Outcome
Michael Faraday	1830s–1840s	Cutting and subdividing permanent magnets; studying field lines	Each fragment formed a new dipole (N and S); no isolated poles observed
Wilhelm Eduard Weber	1830s–1850s	Magnetization/demagnetization experiments; current-based magnetization models	Magnetism behaved as dipoles; no evidence of single poles
James Clerk Maxwell	1860s–1870s	Theoretical synthesis + experiments on magnetization; formulation of field theory	Established $\nabla \cdot \mathbf{B} = 0$ in classical EM $\rightarrow$ no magnetic charges
Pierre Curie	1890s	Heating magnets near Curie temperature; demagnetization studies	Loss/recovery of magnetization but never isolated poles

More Advanced Experiments (Searching for Monopoles)

- The Event: On Sunday, February 14, 1982 (Valentine's Day), a superconducting detector in a basement lab at Stanford registered a sudden, precise change in current. The signal matched the theoretical prediction for a single magnetic monopole—a particle with only a north or south magnetic pole, rather than both—passing through the coil.
- The Apparatus: The experiment utilized a superconducting niobium loop, or SQUID (Superconducting Quantum Interference Device), to monitor for magnetic flux changes.
- The Significance: If confirmed, it would have been one of the most important physics discoveries of the century, providing evidence for Grand Unified Theories (GUTs) which predict the existence of these massive particles from the early universe.
- The Aftermath: Despite building larger and more sensitive detectors and running the experiment for years afterward, Cabrera never detected another signal. The scientific community now largely regards the event as an "ultra-rare glitch" or an unexplained technical artifact, rather than a definitive discovery of a new particle.

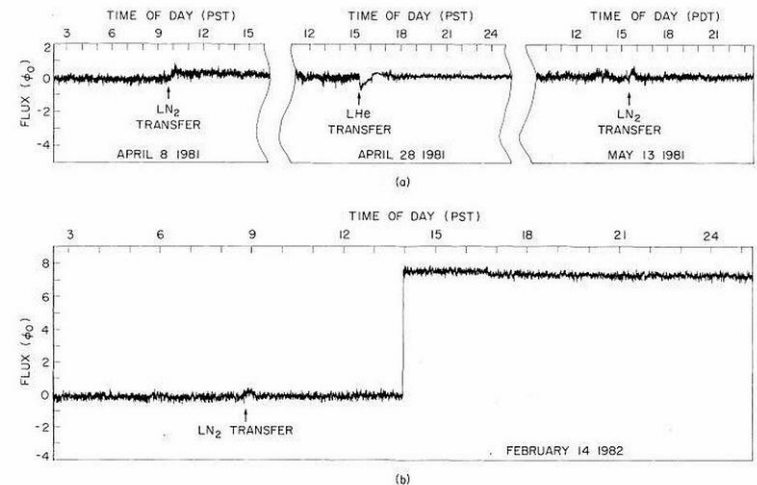
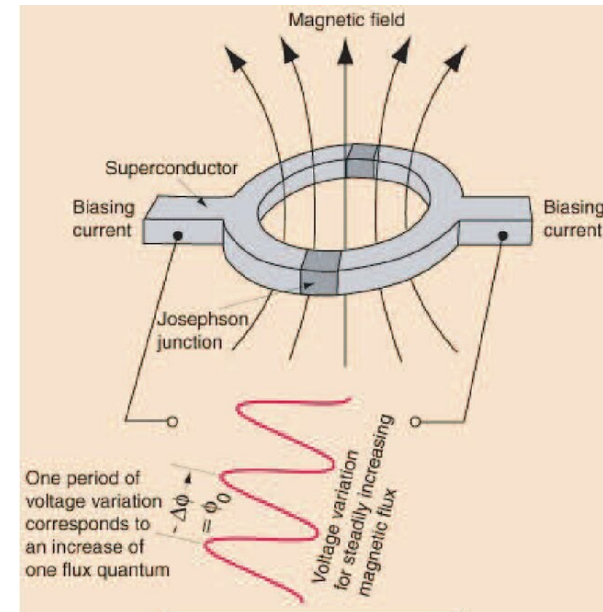


FIG. 2. Data records showing (a) typical stability and (b) the candidate monopole event.

Particle accelerators (modern experiments)

Researchers at the Large Hadron Collider (LHC) are searching for monopoles produced in high-energy collisions. Monopoles would be highly ionising, meaning they would strip electrons from atoms and would leave behind significant energy deposits in detectors. ATLAS physicists previously searched for distinctive monopole signatures using proton-proton collision data from Run 1 (2010-2012) and Run 2 (2015-2018) of the LHC, with no signal found.

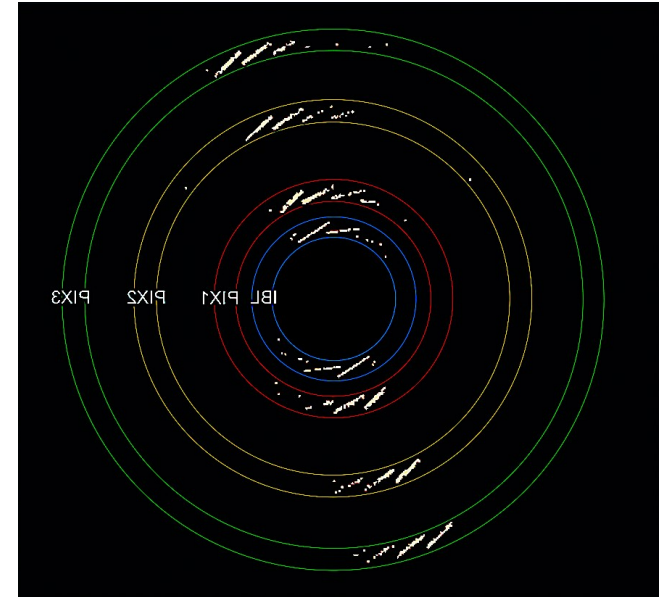


Figure 1: Simulated low-energy monopole-pair interaction with the ATLAS pixel detector in ultraperipheral heavy-ion collisions, with each detector layer region shown as a concentric ring, and with individual Pixel hits shown as orange dots. (Image: ATLAS Collaboration/CERN)

Magnetic Field

- A magnetic field is a vector field that describes the magnetic influence of electrical currents and magnetized materials (Wikipedia)

Intensity of magnetising field (H)

Magnetising Field Intensity or magnetic intensity (H)

The degree to which a magnetic field can magnetize a material is called magnetic field or magnetizing force or Magnetic Intensity.

OR

The number of ampere-turns (NI) per unit length of a solenoid required to produce a given magnetising field

H is measured in Ampere/meter(A/m).

Magnetic Field Intensity (H)

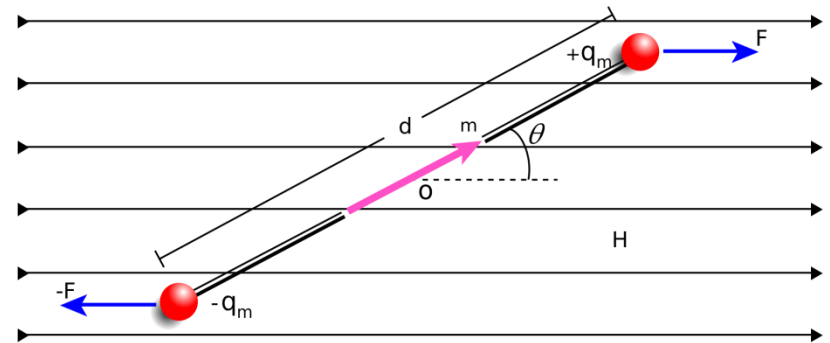
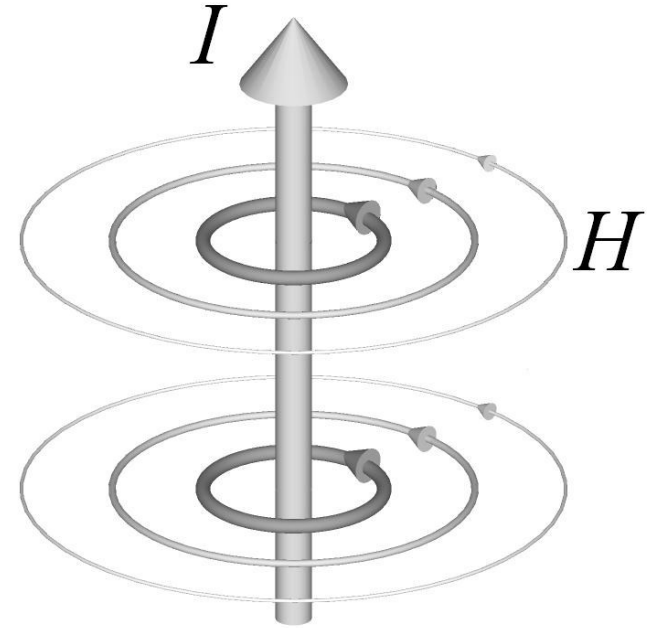
$$\oint H \cdot dl = I$$

H is current/distance [A/m]

$$\tau = \mu_0 [m \times H]$$

τ is the torque acting on a small magnet m in magnetic field H

μ_0 is a constant called the vacuum permeability $\mu_0 = 4\pi \times 10^{-7} \text{ V} \cdot \text{s}/(\text{A} \cdot \text{m})$

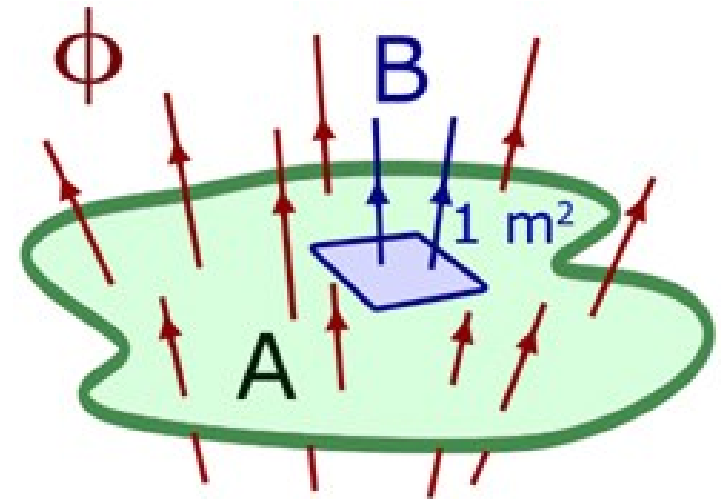


Magnetic Flux Density

Flux density (Magnetic flux density) **B**. The flux density is the number of magnetic lines of flux that pass through a certain point on a surface. The SI unit is T (tesla), which is weber per square meter (Wb/m²) and the unit in the CGS system is G (gauss). 1 tesla is equivalent to 10,000 gauss.

The Magnetic Flux Density (**B**) is related to the Magnetic Field Intensity (**H**) by:

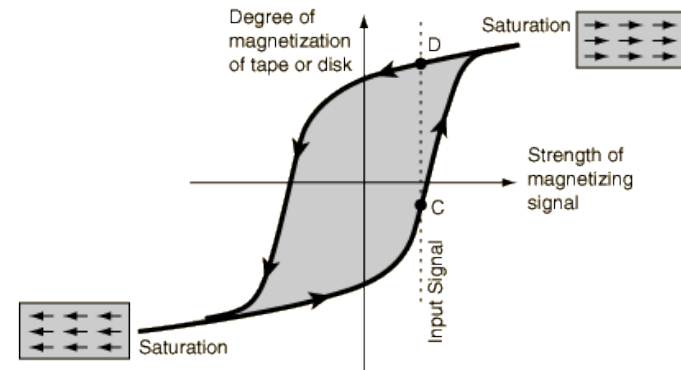
$$\vec{B} = \mu_0 \cdot \mu_r \cdot \vec{H} \quad [\text{Wb/m}^2]$$



μ_0 is a constant called the vacuum permeability $\mu_0 = 4\pi \times 10^{-7} \text{ V} \cdot \text{s}/(\text{A} \cdot \text{m})$

μ_r is the relative permeability characterizing material

Unlike ϵ_r , μ_r is not a constant but may vary significant with the applied magnetic field



Gauss' Law for Magnetic Field

closed surface integral

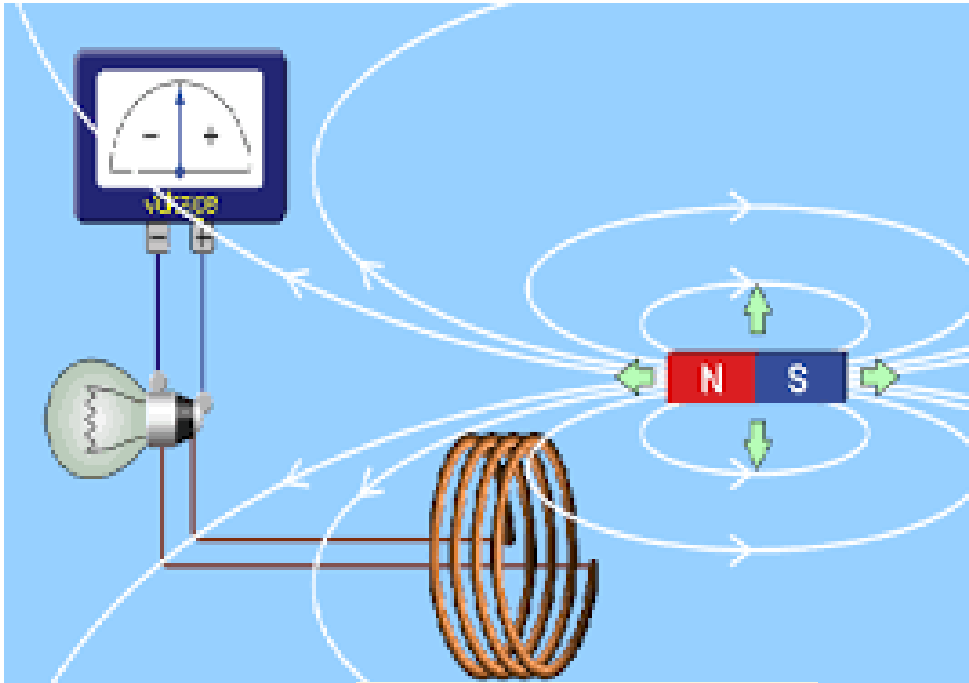
$$\oint_S \vec{B} \cdot d\vec{S} = 0$$

Total magnetic flux through the closed surface should be equal to zero

It does not mean that the magnetic field is zero everywhere!

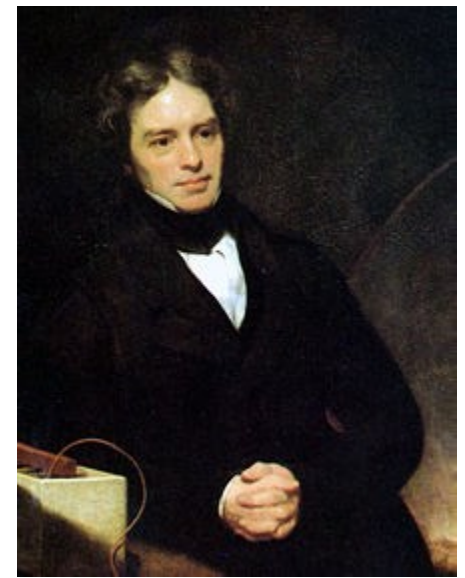


Faraday's Law

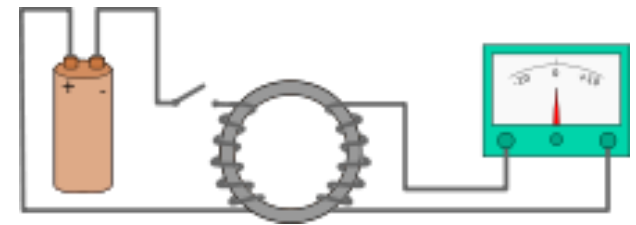


$$\mathcal{E} = - \frac{\Delta \Phi_B}{\Delta t}$$

Faraday's law of induction (shortly called Faraday's law throughout this document) is a basic law of electromagnetism predicting how a magnetic field will interact with an electric circuit to produce an electromotive force (EMF)—a phenomenon called electromagnetic induction. It is the fundamental operating principle of transformers, inductors, and many types of electrical motors, generators and solenoids.



Michael Faraday (22 September 1791
– 25 August 1867)



Faraday's Law of Induction; Lenz's Law

The induced emf in a wire loop is proportional to the rate of change of magnetic flux through the loop.

Magnetic flux: $\Phi_B = B_{\perp} A = BA \cos \theta$ (on formula sheet)

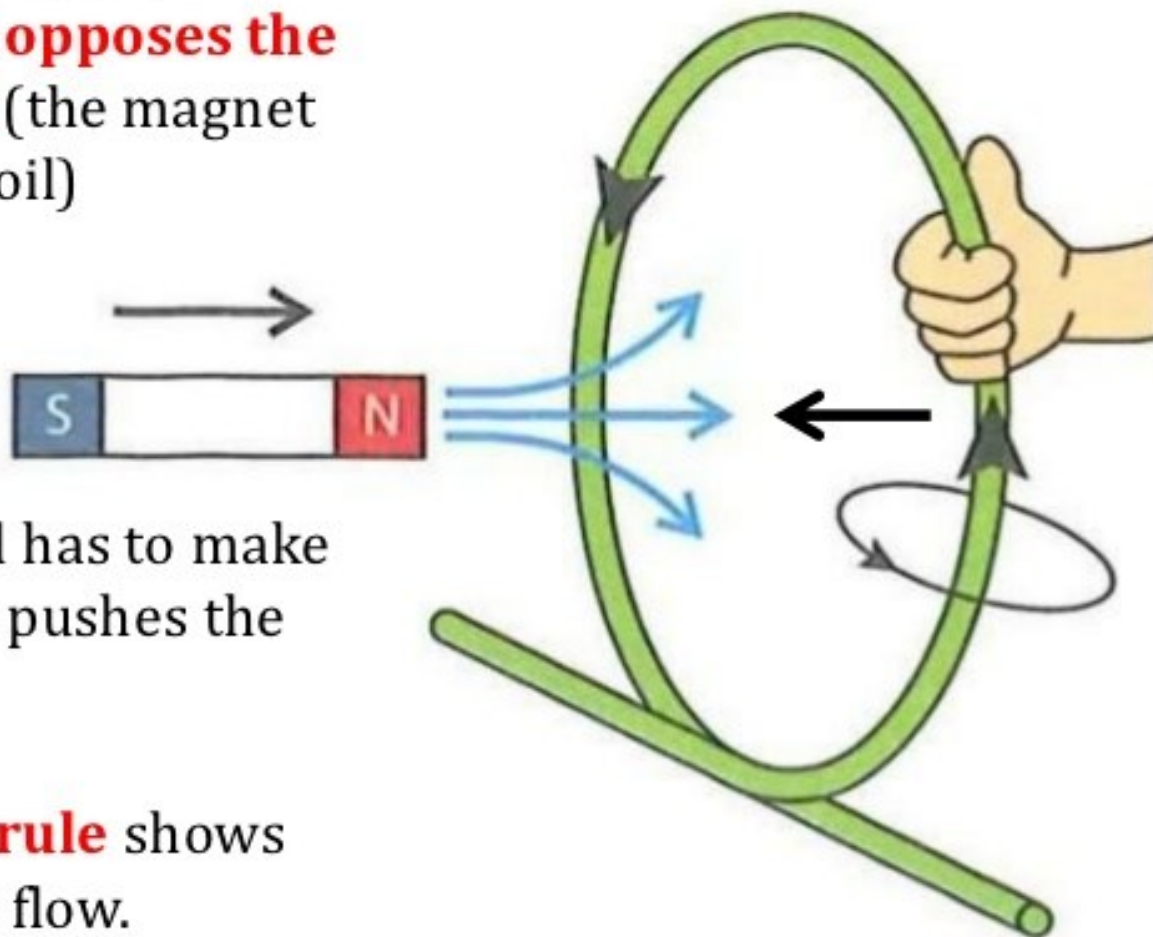
Unit of magnetic flux: weber, Wb.

$$1 \text{ Wb} = 1 \text{ T} \cdot \text{m}^2$$

Faraday's and Lenz's Laws – Example 1

B) Lenz's Law

The direction of the Induced current is such that **it opposes the change producing it** (the magnet moving towards the coil)



The current in the coil has to make a **magnetic field** that pushes the Magnet away....

The **Right hand grip rule** shows how the current must flow.

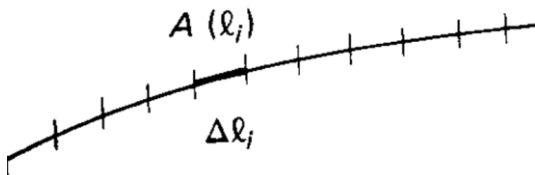
Faraday's Law

$$\oint_c \mathbf{E} \cdot d\boldsymbol{\ell} = - \frac{d}{dt} \int_s \mathbf{B} \cdot d\mathbf{s}$$

Lenz's Law

closed contour integral

surface (enclosed by the contour) Integral



$$\int_c A(\ell) d\ell = \lim_{\substack{\Delta \ell_i \rightarrow 0 \\ N \rightarrow \infty}}$$

$$\sum_{i=1}^N A(\ell_i) \Delta \ell_i$$



Example: calculating work done

$$\Delta W_i = q \underbrace{E_i \cos \alpha_i}_{\substack{\text{component} \\ \text{of force} \\ \text{along the} \\ \text{the contour}}} \Delta \ell_i$$

The total work done W in moving the charge q is then given by

$$\begin{aligned} W &= q \sum_{i=1}^N E_i \cos \alpha_i \Delta \ell_i \\ &= q \sum_{i=1}^N \mathbf{E}_i \cdot \mathbf{t} \Delta \ell_i \end{aligned}$$

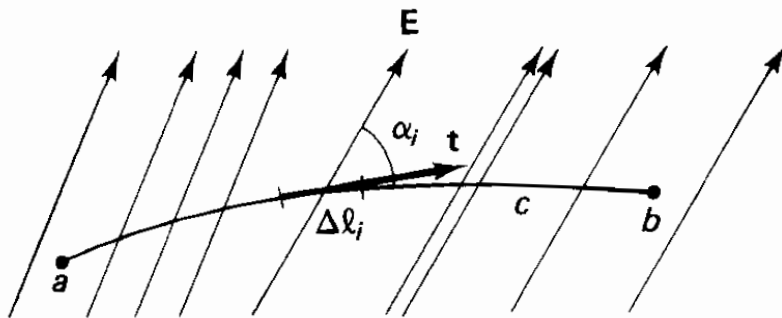


Figure 1.38 A charge q is moving along the contour c in the electric field.

Different types of EMF

1. A time-varying magnetic field linking a stationary loop; the induced emf is then called the *transformer emf*, $V_{\text{emf}}^{\text{tr}}$.
2. A moving loop with a time-varying surface area (relative to the normal component of $\mathbf{B}$) in a static field $\mathbf{B}$; the induced emf is then called the *motional emf*, $V_{\text{emf}}^{\text{m}}$.
3. A moving loop in a time-varying field $\mathbf{B}$.

The total emf is given by

$$V_{\text{emf}} = V_{\text{emf}}^{\text{tr}} + V_{\text{emf}}^{\text{m}},$$



Stationary Loop in Time-Varying $\mathbf{B}$

It is important to remember that $\mathbf{B}_{\text{ind}}$ serves to oppose the change in $\mathbf{B}(t)$, and not necessarily $\mathbf{B}(t)$ itself.

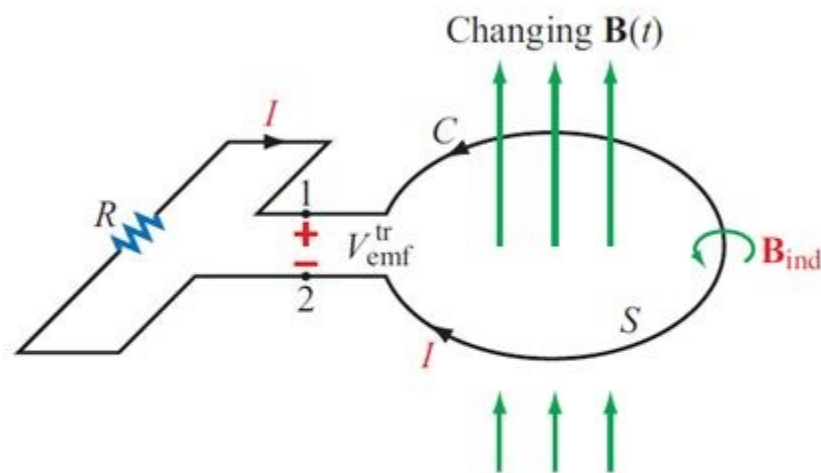
$$V_{\text{emf}}^{\text{tr}} = -N \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s} \quad (\text{transformer emf}),$$

The connection between the direction of $d\mathbf{s}$ and the polarity of $V_{\text{emf}}^{\text{tr}}$ is governed by the following right-hand rule: if $d\mathbf{s}$ points along the thumb of the right hand, then the direction of the contour C indicated by the four fingers is such that it always passes across the opening from the positive terminal of $V_{\text{emf}}^{\text{tr}}$ to the negative terminal.

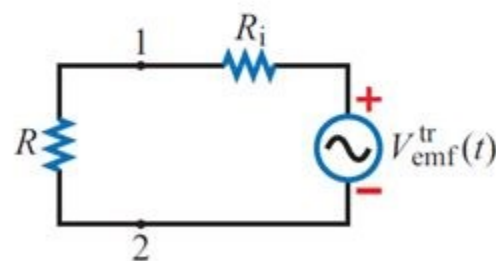
$$I = \frac{V_{\text{emf}}^{\text{tr}}}{R + R_i}. \quad (6.9)$$

For good conductors, R_i usually is very small, and it may be ignored in comparison with practical values of R .

*The polarity of $V_{\text{emf}}^{\text{tr}}$ and hence the direction of I is governed by **Lenz's law**, which states that the current in the loop is always in a direction that opposes the change of magnetic flux $\Phi(t)$ that produced I .*



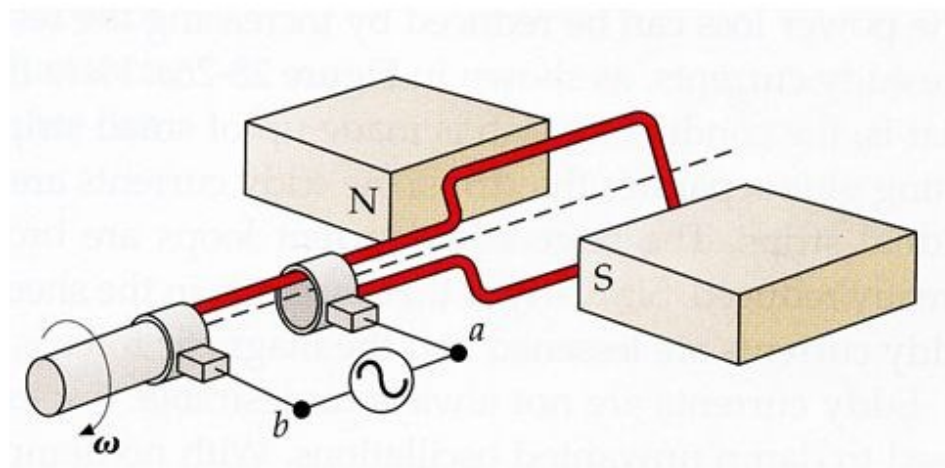
(a) Loop in a changing $\mathbf{B}$ field



(b) Equivalent circuit

Figure 6-2: (a) Stationary circular loop in a changing magnetic field $\mathbf{B}(t)$, and (b) its equivalent circuit.

Electric Generators (AC)

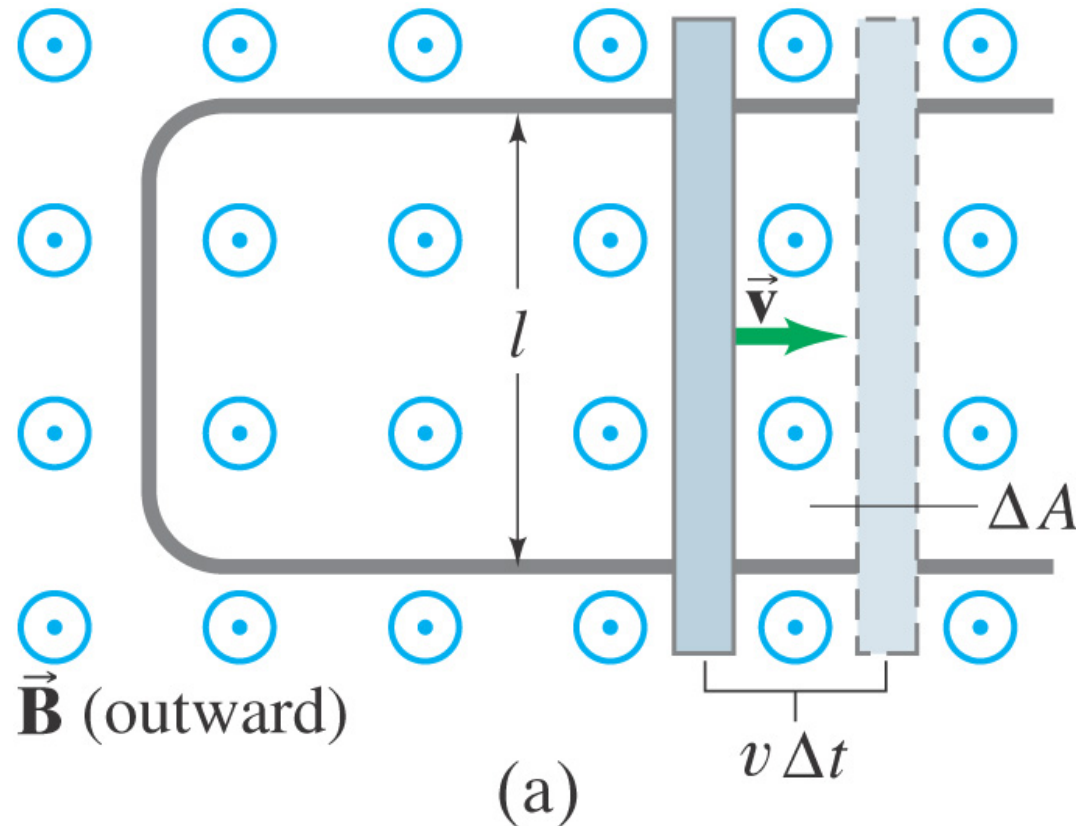


- A mechanically turned armature, consisting of N-loops of a coil wound on the armature
- As the armature is rotated, an EMF is induced in the rotating coil, which alternates as the magnetic flux through the loop reverses. The induced EMF is:

$$E = NBA \sin(\omega t) \text{ or } E_0 \sin(\omega t) \text{ where } E_0 = NBA$$

21.3 EMF Induced in a Moving Conductor

This image shows another way the magnetic flux can change:



Ampere's Law

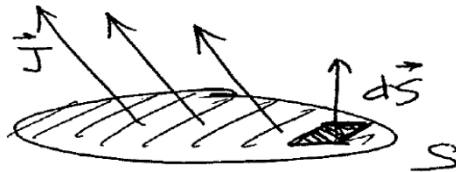
The line integral of the magnetic flux density along a closed contour C is equal to the total current crossing the area S enclosed by this contour.

$$\oint_C \frac{\vec{B}}{\mu_0} \cdot d\vec{C} = \int_S \vec{J} \cdot d\vec{S} + \frac{d}{dt} \int_S \epsilon_0 \vec{E} \cdot d\vec{S}$$

$\int_S \vec{J} \cdot d\vec{S}$ - current resulting from flow of charges

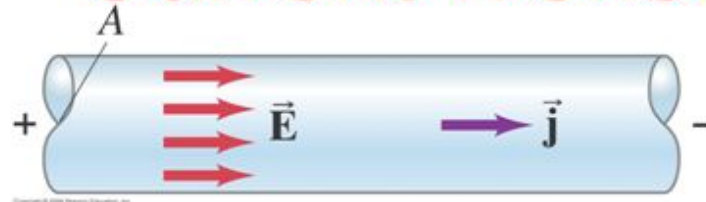
$\vec{J}$ - current flux density

$$\vec{J} = \rho \cdot \vec{V}$$



$\frac{d}{dt} \int_S \epsilon_0 \vec{E} \cdot d\vec{S}$ - "displacement" current (added by Maxwell)

Current Density



- When a potential difference is applied across a conducting wire, an electric field is generated parallel to the walls of the wire
- Inside the conductor, the E-field is no longer zero, because charges are free to move within the conductor
- Current Density is defined as the current through the wire per unit of Cross-Sectional Area

$$j = \frac{I}{A} \text{ or } I = j \cdot A$$

$$[\text{A/m}^2]$$

- If the current density is not uniform:

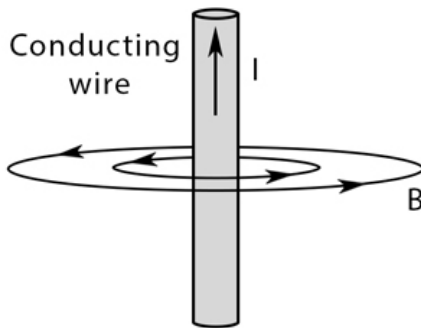
$$I = \int \vec{j} \cdot d\vec{A}$$

- The direction of j is usually in the direction of the E-Field

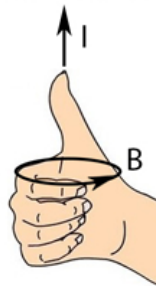
Ampere's Law

$$\oint_C \frac{\vec{B}}{\mu_0} \cdot d\vec{\ell} = \int_S \vec{J} \cdot d\vec{S} + \frac{d}{dt} \int_S \epsilon_0 \vec{E} \cdot d\vec{S}$$

Ampere's Law



Right hand thumb rule



Integral form: $\oint \vec{B} \cdot d\vec{\ell} = \mu_0 I$

Differential form: $\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}$

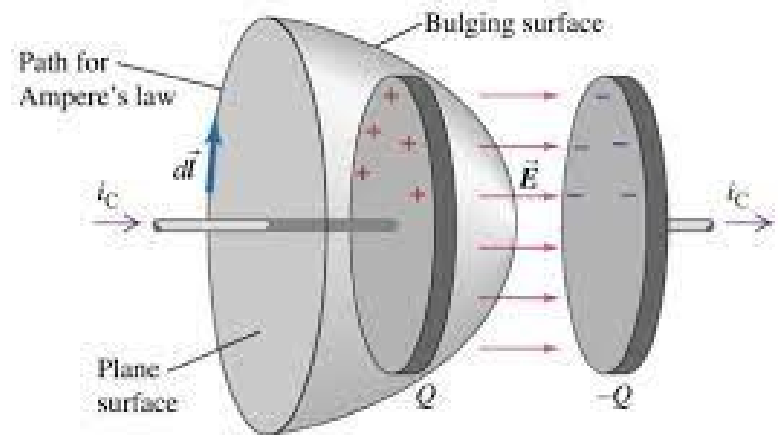
I : Electric current

B : Magnetic field

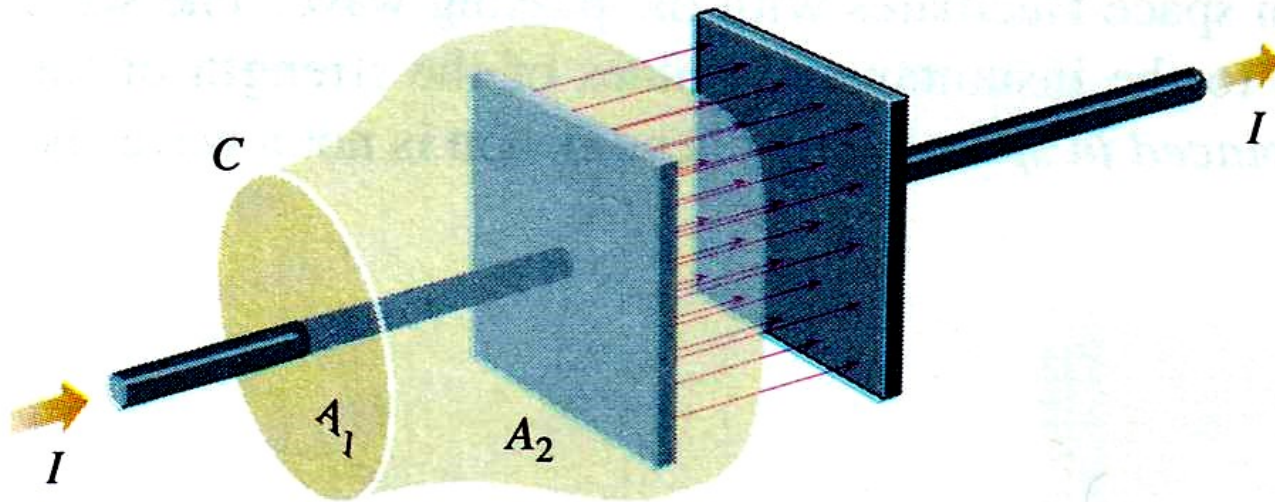
μ_0 : Permeability of free space

J : Current density

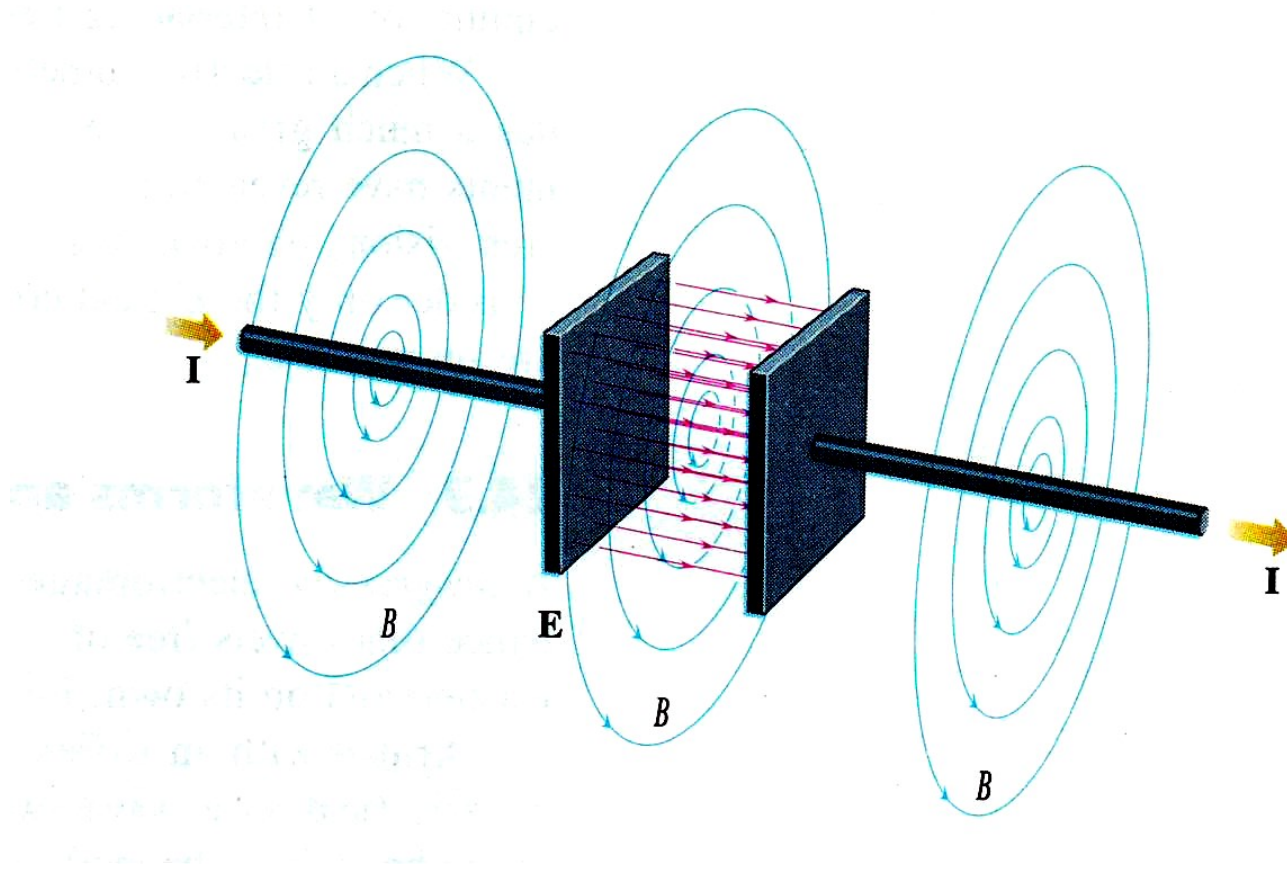
Thumb points in the direction of the electric current and fingers curl around the current indicating the direction of the magnetic field



Imagine a wire connected to a charging or discharging capacitor. The area in the Amperian loop could be stretched into the open region of the capacitor. In this case there would be current passing through the loop, but not through the area bounded by the loop.

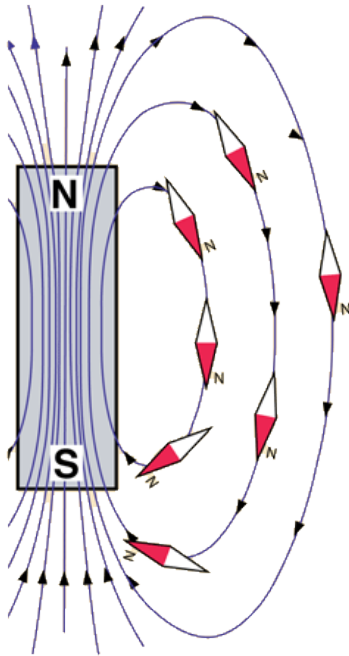


If Ampere's Law still holds, there must be a magnetic field generated by the changing E-field between the plates. This induced B-field makes it look like there is a current (call it the **displacement current**) passing through the plates.

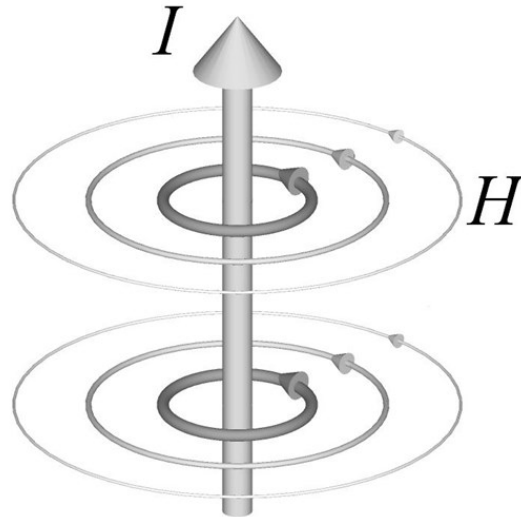


Magnetic Field Lines

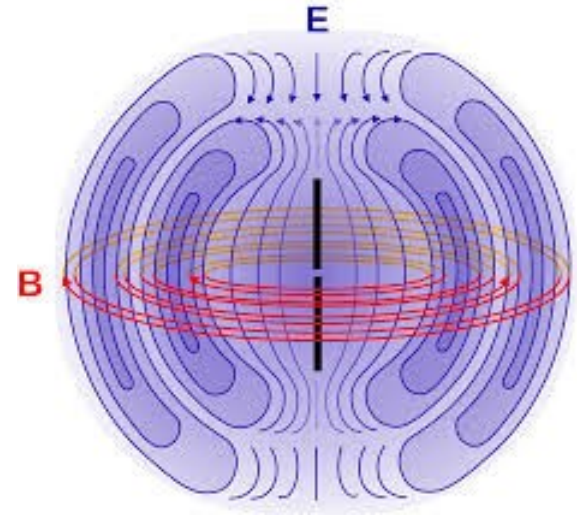
Permanent magnet



Electric current



Time-varying electric field



Mapping the magnetic field of an object is simple in principle. First, measure the strength and direction of the magnetic field at a large number of locations (or at every point in space). Then, mark each location with an arrow (called a vector) pointing in the direction of the local magnetic field with its magnitude proportional to the strength of the magnetic field.

An alternative method to map the magnetic field is to 'connect' the arrows to form magnetic field lines. The direction of the magnetic field at any point is parallel to the direction of nearby field lines, and the local density of field lines can be made proportional to its strength. Magnetic field lines are like streamlines in fluid flow, in that they represent something continuous, and a different resolution would show more or fewer lines.

An advantage of using magnetic field lines as a representation is that many laws of magnetism (and electromagnetism) can be stated completely and concisely using simple concepts such as the 'number' of field lines through a surface. These concepts can be quickly 'translated' to their mathematical form. For example, the number of field lines through a given surface is the surface integral of the magnetic field. (Wikipedia)

Lorentz Force Equation

Lorentz force, the force exerted on a charged particle q moving with velocity $\mathbf{v}$ through an electric field $\mathbf{E}$ and magnetic field $\mathbf{B}$. The entire electromagnetic force $\mathbf{F}$ on the charged particle is called the Lorentz force (after the Dutch physicist Hendrik A. Lorentz) and is given by $\mathbf{F} = q\mathbf{E} + q\mathbf{v} \times \mathbf{B}$.

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

$\mathbf{F}$ = force

q = electric charge

$\mathbf{E}$ = external electric field

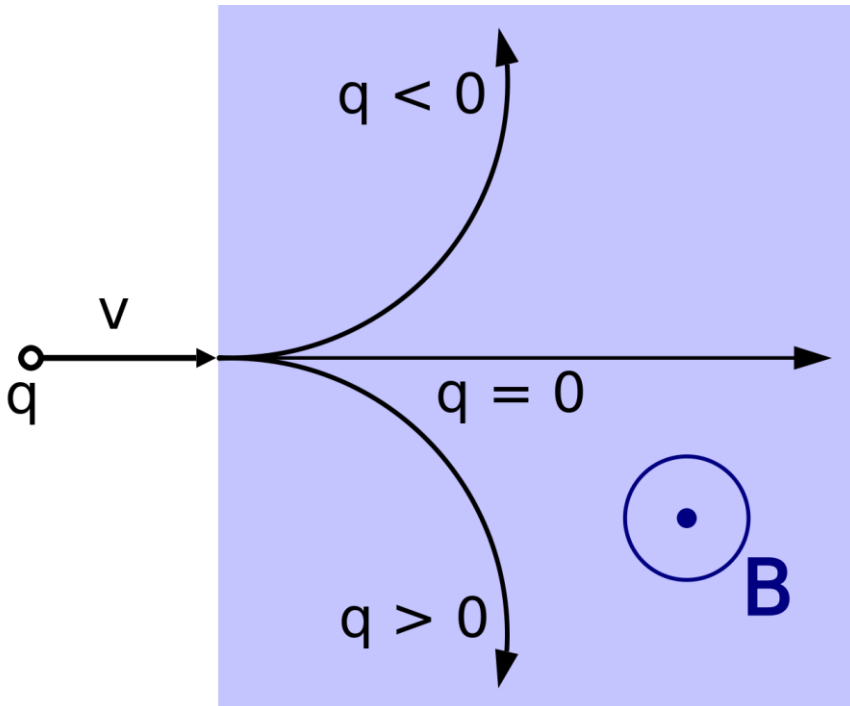
$\mathbf{v}$ = velocity

$\mathbf{B}$ = magnetic field

Charged Particle Trajectory

$$\vec{E} = 0$$

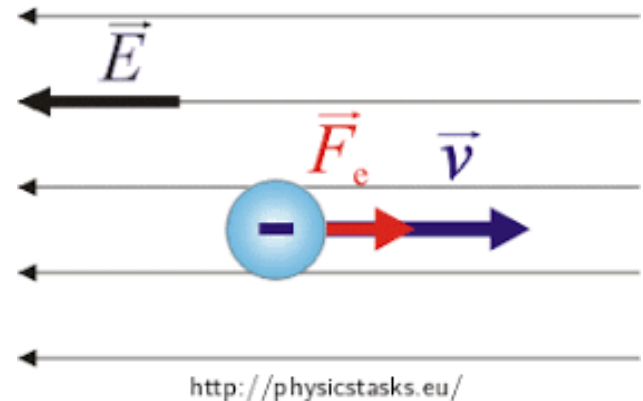
$$\vec{F} = q(\vec{v} \times \vec{B})$$



Trajectory of a particle with a positive or negative charge q under the influence of a magnetic field $\vec{B}$, which is directed perpendicularly out of the screen.

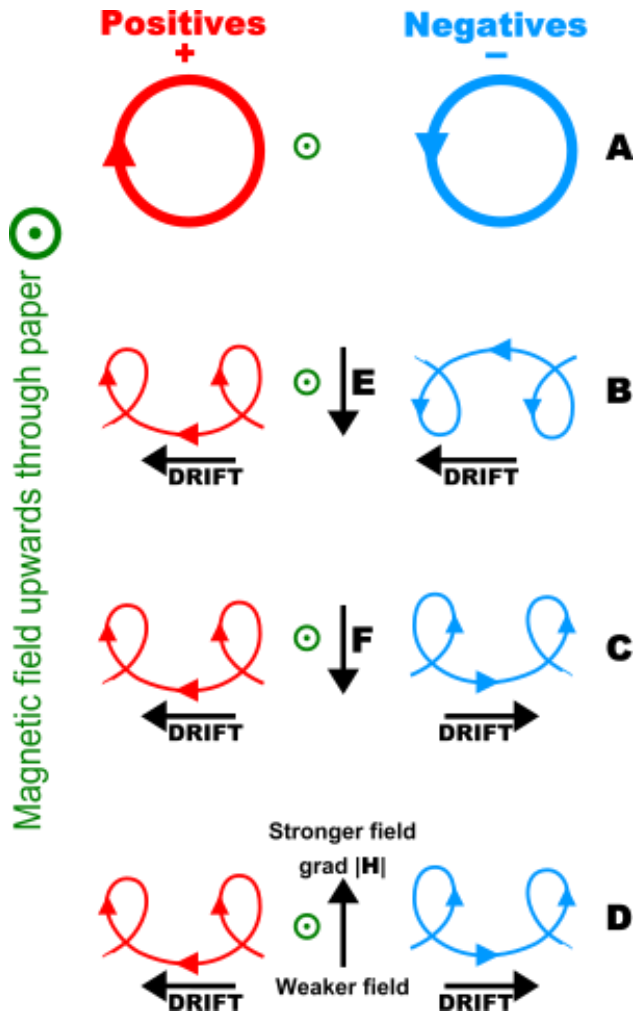
$$\vec{B} = 0$$

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$$



Trajectory of a particle with a positive or negative charge q under the influence of an electric field $\vec{E}$, which is directed in-place of the screen.

Charged Particle Trajectory

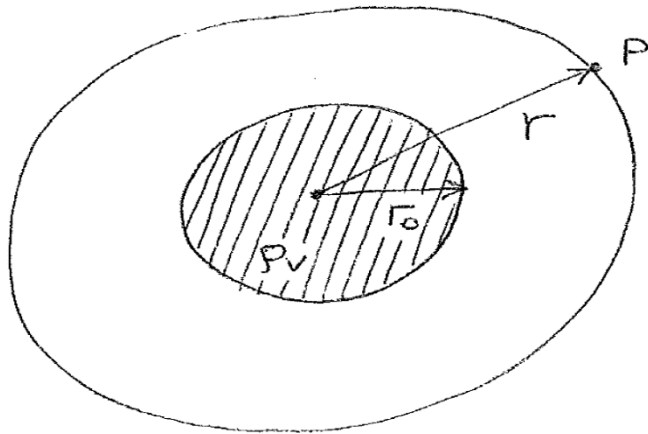


Beam of electrons moving in a circle, due to the presence of a magnetic field. Purple light revealing the electron's path in this [Teltron tube](#) is created by the electrons colliding with gas molecules [Wiki].

Charged particle drifts in a homogeneous magnetic field. (A) No disturbing force (B) With an electric field, E (C) With an independent force, F (e.g. gravity) (D) In an inhomogeneous magnetic field, $\text{grad } H$

Classical Example

ρ_v - charge density



$$\oint_S \epsilon_0 \vec{E} \cdot d\vec{S} = \int_V \rho_v dV$$

case: $r > r_0$

$$\text{Total flux} = \oint_S \epsilon_0 E_r \vec{a}_r \cdot d\vec{S}_r$$

We need element of area $d\vec{S}_r$
[we look page 15]

$$\vec{S}_r = r^2 \sin \theta d\theta d\phi \vec{a}_r$$

$$\begin{aligned} \theta &\in (0; \pi) \\ \phi &\in (0; 2\pi) \end{aligned}$$

$$\text{Total flux} = \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} \epsilon_0 E_r \vec{a}_r \cdot r^2 \sin \theta d\theta d\phi \vec{a}_r =$$

$$\therefore \text{Total Flux} = \epsilon_0 E_r r^2 \int_{\phi=0}^{2\pi} d\phi \int_{\theta=0}^{\pi} \sin\theta d\theta =$$

$$= 2\pi \epsilon_0 E_r r^2 (-\cos\theta) \Big|_0^{\pi} = 4\pi \epsilon_0 E_r \cdot r^2$$

Charge enclosed by the surface :

$$Q = \int_V \rho_v dV \quad [\text{We need element of volume — look page 15}]$$

$$dV = r^2 \sin\theta dr d\theta d\phi$$

$$Q = \int_V \rho_v \cdot r^2 \sin\theta dr d\theta d\phi =$$

$$= \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} \int_{r=0}^{r_0} \rho_v r^2 \sin\theta dr d\theta d\phi =$$

$$= 2\pi \cdot 2 \cdot \rho_v \cdot \frac{r_0^3}{3} = \frac{4}{3} \pi \rho_v r_0^3$$

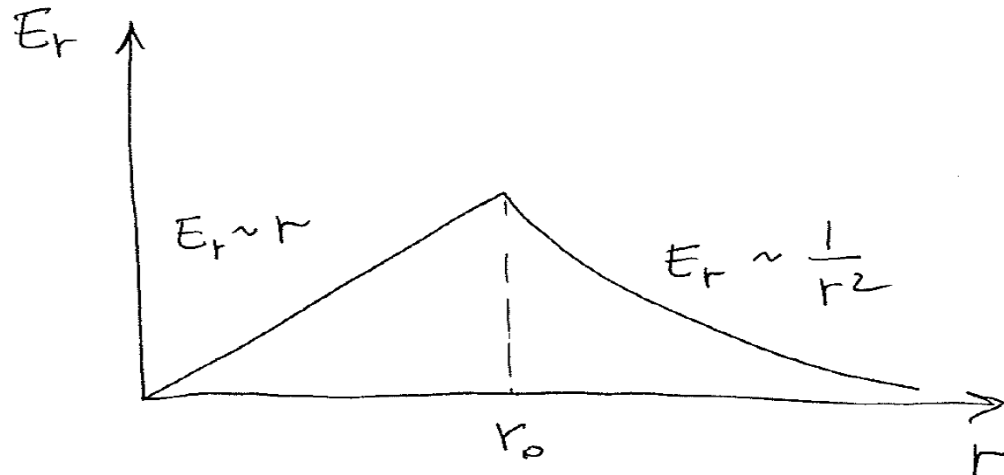
$$4\pi\epsilon_0 E_r \cdot r^2 = \frac{4}{3}\pi\rho_v \cdot r^3$$

$$\Rightarrow \boxed{\begin{aligned} E_r &= \frac{\rho_v r}{3\epsilon_0} \quad (\text{V/m}) \quad r < r_0 \\ \vec{E} &= E_r \vec{a}_r \end{aligned}} \quad (2)$$

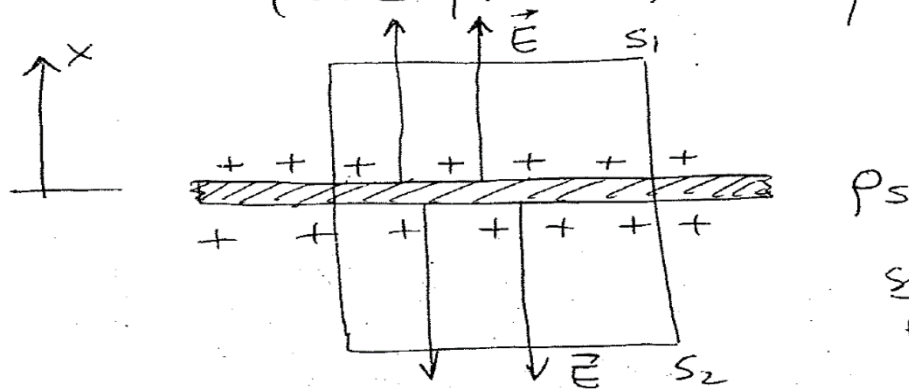
Check boundary:

$$E_{r=r_0} = \frac{\rho_v r_0}{3\epsilon_0} \quad (\text{from (2)})$$

$$\text{From (1): } E_{r=r_0} = \frac{\rho_v r_0^3}{3\epsilon_0 r_0^2} = \frac{\rho_v r_0}{3\epsilon_0} \quad !!!$$



Infinitely Large Plane Surface (one plate of a capacitor)



surface: rectangular
parallelepiped

$$\vec{E}_{\text{front}} = E_x \cdot \vec{a}_x$$

$$\oint_S \epsilon_0 \vec{E} \cdot d\vec{S} = Q$$

$$\int_{S_1} \epsilon_0 E_x \vec{a}_x \cdot A \vec{a}_x +$$

$$+ \int_{S_2} \epsilon_0 (-E_x \vec{a}_x) \cdot (-A \vec{a}_x) = Q$$

$$2\epsilon_0 E_x \cdot A = \rho_s \cdot A$$

$$\Rightarrow E_x = \frac{\rho_s}{2\epsilon_0}$$

$$\vec{E} = E_x \vec{a}_x = \frac{\rho_s}{2\epsilon_0} \vec{a}_x$$

for $x > 0$

$$\vec{E} = -\frac{\rho_s}{2\epsilon_0} \vec{a}_x, \quad x < 0$$

Homework 1 has been posted

The due date is April 15

26. Consider the contour c shown in Figure P1.26 and the vector field

$$\mathbf{F} = 2\rho(z^2 + 1) \cos \phi \mathbf{a}_\rho - \rho(z^2 + 1) \sin \phi \mathbf{a}_\phi + 2\rho^2 z \cos \phi \mathbf{a}_z.$$

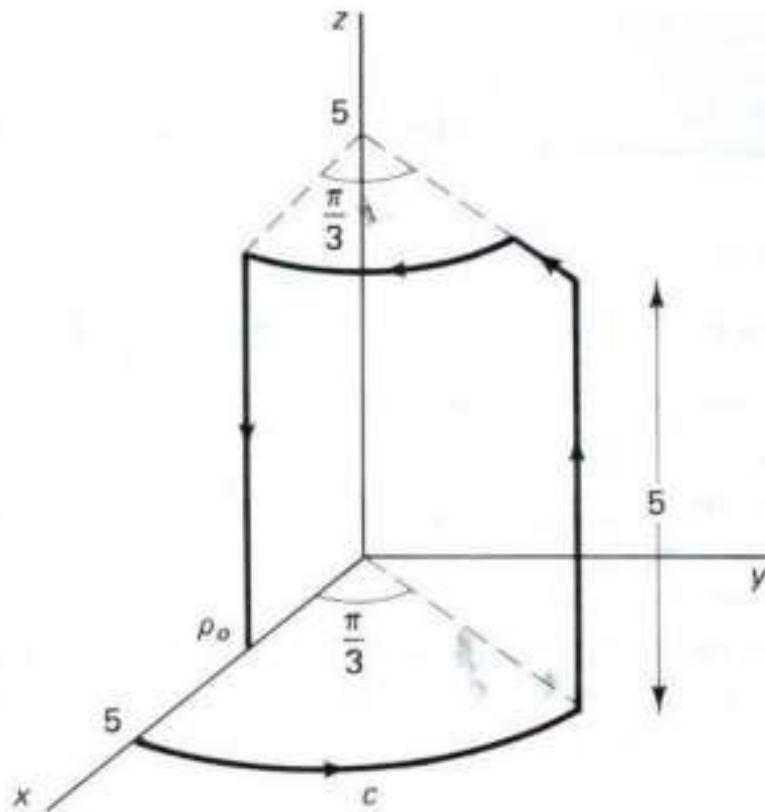


Figure P1.26

There is no late submission.